



Part II: Wyckoff Stock Analysis Module

Complete Technical Documentation & Methodology Guide



What is Part II (Stock Analysis)?

Part II of the Wyckoff Trader project is the **Stock Analysis Module** that:

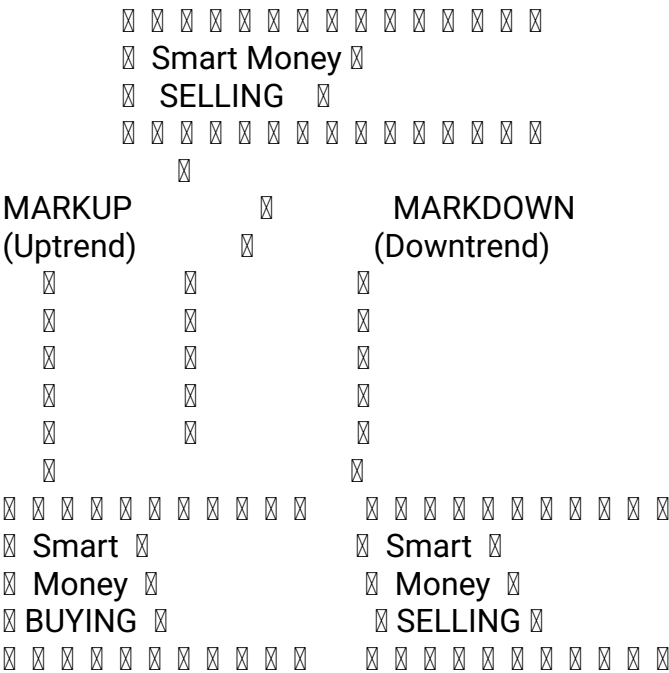
- 1. Fetches real-time stock data using yfinance API
- 2. Detects Wyckoff events (SC, BC, SP, UT, SOS, SOW, LPS, LPSY)
- 3. Identifies the current Wyckoff phase (A, B, C, D, E)
- 4. Determines market bias (Bullish, Bearish, Neutral)
- 5. Runs backtesting based on Wyckoff trading rules
- 6. Visualizes everything with interactive Plotly charts



The Wyckoff Price Cycle

Richard Wyckoff identified that markets move in a **repeating cycle** controlled by large institutional operators (the "Composite Man"):

DISTRIBUTION



ACCUMULATION MARKDOWN

The Four Market Phases:

Phase	Description	Who's in Control
Accumulation	Smart money quietly buying	Institutions (Strong Hands)
Markup	Price rising, public joins	Demand > Supply
Distribution	Smart money quietly selling	Institutions exiting
Markdown	Price falling, public panics	Supply > Demand



Wyckoff's Three Fundamental Laws

Law 1: Supply and Demand

SUPPLY AND DEMAND LAW

When DEMAND > SUPPLY Price RISES

When SUPPLY > DEMAND Price FALLS

When SUPPLY = DEMAND Price RANGES

This is the MOST fundamental law of price movement.

Everything else in Wyckoff analysis builds on this.

How we detect it: Compare buying volume vs selling volume, analyze price spread relative to volume.

Law 2: Cause and Effect

CAUSE AND EFFECT LAW

CAUSE (Trading Range) EFFECT (Price Movement)

ACCUMULATION = MARKUP

(Building Cause) (The Effect)

Time in range Size of move

Longer accumulation = Larger subsequent move

☐
 ☐ Characteristics:
 ☐ • Moderate increase in volume
 ☐ • Price temporarily holds, then continues down
 ☐ • Early warning that downtrend is weakening
 ☐
 ☐ Chart Pattern:
 ☐
 ☐
 ☐ \
 ☐ \ ☐ Downtrend
 ☐ \
 ☐ \ ☐ PS: Small bounce
 ☐ \
 ☐ \ ☐ Continues down to SC
 ☐
 ☐ Trading Action: Do NOT buy yet. Just observe.

2. SC - Selling Climax ☆ (Key Event)

☐
 ☐ SELLING CLIMAX (SC)
 ☐
 ☐
 ☐ Definition: Panic selling that exhausts supply and stops
 ☐ the downtrend. Smart money starts buying here.
 ☐
 ☐ Detection Rules (our code):
 ☐ • Volume Ratio > 2.0 (very high volume spike)
 ☐ • Price Change < -3% (significant drop)
 ☐ • Close > Low + 30% of range (closes off the lows)
 ☐
 ☐ Chart Pattern:
 ☐
 ☐ \
 ☐ \
 ☐ \
 ☐ \
 ☐ \
 ☐ \ ☐ SC: Huge volume, wide spread
 ☐ \ ☐ Volume spike
 ☐ \ ☐ Price recovers from lows
 ☐
 ☐ What's Happening:
 ☐ • Retail traders panic selling
 ☐ • Institutions absorbing all the selling
 ☐ • Transfer from "weak hands" to "strong hands"
 ☐
 ☐ Trading Action: Watch closely, but don't buy yet.

3. AR - Automatic Rally

AUTOMATIC RALLY (AR)
 Definition: The natural bounce after Selling Climax.
 Occurs because selling pressure is temporarily exhausted.
 Characteristics:
 • Follows immediately after SC
 • Short covering + new buying
 • Sets the upper boundary of trading range
 Chart Pattern:
 AR: Sets resistance
 SC
 Trading Range established
 Trading Action: Note the AR high as resistance level.

4. ST - Secondary Test

SECONDARY TEST (ST)
 Definition: Price returns to SC area to test if selling pressure has truly been absorbed.
 Successful ST characteristics:
 • LESS volume than SC
 • NARROWER price spread than SC
 • Stops at or above SC low
 Chart Pattern:
 AR
 ST Less volume, higher low = BULLISH
 SC
 Volume comparison
 What it tells us:
 • If ST shows less selling = Accumulation confirmed
 • If ST breaks below SC = More weakness, not ready yet
 Trading Action: Confirms accumulation, but still wait.

5. SP - Spring ☆☆☆ (Most Important Entry Signal!)

SPRING (SP) - THE KEY ENTRY SIGNAL

Definition: A false breakdown below support that traps sellers and provides the best entry opportunity.

Detection Rules (our code):

- Low breaks below support level
- Close recovers ABOVE support
- Volume Ratio < 1.5 (LOW volume = lack of real selling)

The Three Types of Springs:

Type 1 (Weak): Type 2 (Moderate): Type 3 (Strong):

Deep penetration Moderate penetration Shallow penetration

Higher volume Lower volume Lowest volume

Needs more testing Good entry BEST entry

Chart Pattern:

Support Level

SPRING: Shakes out weak holders

LOW VOLUME = Real buyers absorb

Why Springs Work:

- Triggers stop losses of early bulls
- Traps new short sellers
- Smart money buys from panicked sellers
- Creates fuel for subsequent rally

Trading Action: PRIMARY BUY SIGNAL! Enter long positions.

6. SOS - Sign of Strength ☆ (Confirmation Signal)

SIGN OF STRENGTH (SOS)

Definition: A rally that breaks above resistance on INCREASING volume, confirming accumulation is complete.

Detection Rules (our code):

- Close breaks above resistance
- Price change > 1.5%

- Volume Ratio > 1.3 (above average volume)

Chart Pattern:

What it tells us:

- Demand has overcome supply
- Accumulation is complete
- Markup phase is beginning

Trading Action: Add to long positions or enter if missed Spring

7. LPS - Last Point of Support (Re-entry Signal)

LAST POINT OF SUPPORT (LPS)

Definition: A pullback after SOS on LOW volume that provides another low-risk entry opportunity.

Detection Rules (our code):

- Recent SOS in last 10 bars
- Pullback (negative return)
- Volume Ratio < 0.8 (below average)
- Stays above support

Chart Pattern:

Why LPS is a great entry:

- Confirms demand is in control
- Lower risk than buying at highs
- Stop loss can be placed below LPS

Trading Action: Excellent place to add or initiate positions

DISTRIBUTION Events (Bearish)

1. BC - Buying Climax

BUYING CLIMAX (BC)

Definition: Euphoric buying that exhausts demand and marks the end of an uptrend. Smart money starts selling here.

Detection Rules (our code):

- Volume Ratio > 2.0 (very high volume)
- Price Change > 3% (significant rise)
- Price near recent highs

Chart Pattern:

BC Huge volume, wide spread at top

What's Happening:

- Retail traders buying in euphoria (FOMO)
- Institutions selling to retail
- Transfer from "strong hands" to "weak hands"

Trading Action: Do NOT buy. Consider taking profits.

2. UT - Upthrust ☆☆☆ (Key Short Signal)

UPTHRUST (UT) - KEY DISTRIBUTION SIGNAL

Definition: A false breakout above resistance that traps buyers and provides a short entry opportunity.

Detection Rules (our code):

- High breaks above resistance
- Close falls BELOW resistance
- Volume Ratio < 1.5 (lack of real buying)

Chart Pattern:

UPTHRUST: False breakout

Resistance (ICE)
LOW VOLUME = Trap

- Why Upthrusts Work (opposite of Spring):
- Triggers buy stops above resistance
 - Traps breakout buyers
 - Smart money sells to trapped buyers
 - Creates selling fuel for markdown

Trading Action: EXIT longs / Enter SHORT positions

3. SOW - Sign of Weakness ☆ (Distribution Confirmation)

SIGN OF WEAKNESS (SOW)

Definition: A decline that breaks below support on INCREASING volume, confirming distribution is complete.

Detection Rules (our code):

- Close breaks below support
- Price change < -1.5%
- Volume Ratio > 1.3

Chart Pattern:

UT
Support (ICE)
SOW breaks below ICE
Volume INCREASING

Trading Action: EXIT all longs. Consider shorts.

4. LPSY - Last Point of Supply

LAST POINT OF SUPPLY (LPSY)

Definition: A weak rally after SOW on LOW volume.
Marks the final opportunity to exit or short.

☐ Trading: NO TRADING - Just observe and establish range ☐

Phase B: Building the Cause

☒ PHASE B: BUILDING THE CAUSE ☒

- ☑ Purpose: Institutions accumulate/distribute while price
- ☑ oscillates within the established trading range

Resistance (Creek/AR)

Price oscillates

Volume diminishes

⊠ What's Happening:

- Multiple tests of support and resistance

- "Cause" is being built for future "Effect"

☒ Trading: Minor trades within range, but risky ☒

Phase C: The Test ☆☆☆ (Most Important!)

☒ PHASE C: THE TEST (Spring/Upthrust) ☒

- ☒ Purpose: Final test to shake out remaining weak hands
- ☒ before the new trend begins

☒ In ACCUMULATION: SPRING

☐ ☐ ☐ ☐ ☐

☒ ☒ ☒ ☒ LOW VOLUME = key! ☒

PHASE E: TREND OUT OF RANGE (Markup/Markdown)

Purpose: The new trend is fully underway

Price leaves the trading range

In MARKUP (after accumulation):

SOS

Former Resistance

LPS LPS

Former Support

Spring

Key Characteristics:

- Higher highs AND higher lows
- Pullbacks are shallow and brief
- Re-accumulation structures may form
- Continues until distribution begins

Trading: Hold positions, trail stops, watch for distribution



Phase Identification Logic (Our Code)

```
def identify_phase(df: pd.DataFrame) -> Dict:
    """
    Identify current Wyckoff phase based on:
    1. Recent events detected
    2. Price trend (30-day change)
    3. Volume trend
    """

    recent = df.tail(30)
    events = recent[recent['event'] != '']['event'].value_counts()
    price_trend = (recent['close'].iloc[-1] / recent['close'].iloc[0] - 1) * 100

    # Decision Logic:

    if 'SC' in events or 'SP' in events:
        if price_trend > 2:
            phase = "Phase C/D - Accumulation Test/Markup"
            bias = "BULLISH"
        else:
```

```
phase = "Phase A/B - Accumulation Building"
bias = "NEUTRAL"
```

```
elif 'BC' in events or 'UT' in events:
    if price_trend < -2:
        phase = "Phase C/D - Distribution Test/Markdown"
        bias = "BEARISH"
    else:
        phase = "Phase A/B - Distribution Building"
        bias = "NEUTRAL"

elif 'SOS' in events:
    phase = "Phase D/E - Markup"
    bias = "BULLISH"

elif 'SOW' in events:
    phase = "Phase D/E - Markdown"
    bias = "BEARISH"

else:
    phase = "Phase B - Trading Range"
    bias = "NEUTRAL"

return {'phase': phase, 'bias': bias}
```

Backtesting Strategy (Our Trading Rules)

Entry Rules

```

ENTRY SIGNALS (GO LONG)

Signal      Condition      Confidence
SPRNG (SP)  False breakdown + recovery  HIGHEST
LPS         Pullback after SOS        HIGH
SOS         Breakout on volume          MEDIUM

Code Logic:
if event in ['SP', 'LPS', 'SOS']:
    position = 'LONG'
    entry_price = current_close

```

Exit Rules

EXIT SIGNALS (CLOSE LONG)

```

# Signal      Condition      Action
# UPTHrust (UT) False breakout above res. EXIT
# LPSY        Weak rally after SOW      EXIT
# SOW         Break below support       EXIT
# STOP LOSS   Price < Entry - 5%        EXIT
# TAKE PROFIT Price > Entry + 15%        EXIT

# Code Logic:
# if event in ['UT', 'LPSY', 'SOW']:
#     exit_trade()
# elif price < entry_price * 0.95: # 5% stop
#     exit_trade(reason='stop_loss')
# elif price > entry_price * 1.15: # 15% profit
#     exit_trade(reason='take_profit')

```

Backtest Metrics

```

# BACKTEST OUTPUT METRICS
# Metric      Description
# Total Trades Number of completed trades
# Win Rate     % of profitable trades
# Total Return Sum of all trade P&Ls
# Average Return Mean P&L per trade
# Max Win      Largest winning trade
# Max Loss     Largest losing trade
# Profit Factor Gross profit / Gross loss
# Avg Holding Days Average trade duration

# Example Output:
# {
#     'total_trades': 5,
#     'win_rate': 60.0,
#     'total_return': 12.5,
#     'avg_return': 2.5,
#     'max_win': 8.2,
#     'max_loss': -5.0,
#     'profit_factor': 2.1
# }

```

When a user selects a stock (e.g., AAPL) and clicks "Analyze", they see:

1. Metric Cards (Top Row)

Price Change +5.2% Total Trades 5 Win Rate 60% Total Return +12.5% Events 8

2. Phase Indicator

Current Market Phase
Phase C/D - Accumulation Test/Markup
Spring detected with upward price action. Watch for Sign of Strength to confirm markup.
Bias: BULLISH

3. Interactive Chart

AAPL - Wyckoff Analysis

Price Chart with:

- Candlesticks (green/red)
- 20-day MA (blue line)
- 50-day MA (purple line)
- Support line (green dashed)
- Resistance line (red dashed)
- Event markers (SC, SP, SOS, etc.)

Volume Panel:

- Volume bars (green up days, red down days)
- Volume MA line

Volume Ratio Panel:

- Volume ratio line (spikes = significant events)
- 2.0 threshold line

4. Event Badges

Detected Wyckoff Events

[SC: 2] [SP: 1] [SOS: 2] [LPS: 3]

5. Trade History

Recent Trades (Backtest)

Entry @ \$178.50 (Spring) Exit @ \$192.30 (Take Profit) | PnL: +7.7%

Entry @ \$185.20 (LPS) Exit @ \$176.00 (Stop Loss) | PnL: -5.0%

Entry @ \$180.40 (SOS) Exit @ \$195.50 (UT) | PnL: +8.4%

🎯 Summary: Part II Requirements Fulfilled

Requirement	Implementation	Status
Fetch real-time stock data	yfinance API	✅
Detect Wyckoff events	Rule-based detection (8 events)	✅
Identify phases	Logic based on events + trend	✅
Determine bias	Bullish/Bearish/Neutral	✅
Backtest strategy	Entry/Exit rules with metrics	✅
Visualize results	Interactive Plotly charts	✅
Modern UI	Dark theme, metric cards, badges	✅

Key Wyckoff Terminology Quick Reference

Term	Definition
Composite Man	The collective actions of smart money/institutions
Trading Range (TR)	Sideways price movement between support and resistance
Creek	Resistance level in accumulation
ICE	Support level in distribution
Spring	False breakdown below support (bullish trap)
Upthrust	False breakout above resistance (bearish trap)
Strong Hands	Institutions, informed traders
Weak Hands	Retail traders, uninformed
Absorption	Large orders absorbing selling/buying pressure
Markup	Uptrend phase after accumulation

Markdown	Downtrend phase after distribution
----------	------------------------------------

*Document Version: 1.0**For GenAI Course - Wyckoff Team*